Gravitational Lens Finding

GSoC 2026 Proposal • ML4SCI / DeepLense • Project DEEPLENSE5

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Synopsis

Strong gravitational lenses are cosmologically rare — approximately 1 in every 1,000 galaxies in a typical wide-field survey. With LSST set to observe billions of galaxies, automated lens finding at scale is an essential enabler for dark matter science. This project delivers a production-grade, physics-informed lens-finding pipeline that combines EfficientNet-based deep learning with domain adaptation techniques and direct LSST RSP integration — addressing the extreme class imbalance (1:100+ lenses to non-lenses) inherent in real survey data.

Motivation & Problem Statement

Current DeepLense lens-finding models are trained exclusively on simulated data. When deployed on real observational images from HSC, DES, or LSST, performance degrades due to domain shift — the mismatch between simulation-rendered PSF, noise, and real telescope atmospheric effects. A production-ready lens finder must solve four intertwined challenges:

- **Class imbalance:** true lenses are 1-in-1000 objects, causing naive classifiers to predict non-lens for everything.
- **Calibrated uncertainty:** downstream spectroscopic follow-up requires probability scores, not hard labels.
- **Physical interpretability:** attention must focus on Einstein ring arcs, not survey artefacts.
- **Domain gap:** models trained on simulations must generalise to real survey images without re-labelling.

Prior Contributions to ML4SCI / DeepLense

I have made eight merged pull requests to the ML4SCI/DeepLense repository, demonstrating sustained, concrete engagement with the codebase prior to GSoC:

PR	Title	Impact
#166	LSST RSP Integration Pipeline	End-to-end RSP catalog → lens finder candidate list
#151	Modular Transform Pipeline	Plug-and-play augmentation for multi-survey training
#148	Dataset Pipeline Refactor	Unified HDF5/FITS loader; cut data loading time 40 %
#144	Bug Fix: Normalisation Mismatch	Corrected per-channel stats; improved val AUC ~0.3 %
#136	HSC Survey Adapter	Added real HSC-SSP image ingestion with WCS handling
#130	WeightedRandomSampler	Balanced batch strategy for 1:100 imbalanced data
#129	Focal Loss Integration	gamma=2 focal loss; improved minority-class recall
#122	Unit Tests for Transforms	Pytest suite covering 14 augmentation operations

Evaluation Test Results

Common Test I — Multi-class Lens Classification

Achieved an ensemble macro-AUC of **0.9979** on the three-class task (no substructure, spherical substructure, vortex substructure).

Macro AUC	AUC no	AUC sphere	AUC vort
0.9979	**0.9978**	**0.9971**	**0.9988**

Specific Test VII — Physics-Informed Neural Network (LensPINN)

I designed and implemented **LensPINN**, a physics-informed architecture that embeds the gravitational lensing equation *beta = theta - alpha(theta)* directly into the network via a differentiable *InverseLensLayer*. The backbone is EfficientNet-B0; the lens model is a Singular Isothermal Sphere (SIS) with a learned

perturbation field (Einstein radius theta_E = 0.3, normalised units). A total-variation regularisation term on the reconstructed source plane enforces physical smoothness — the physics consistency loss weight lambda = 0.01.

Metric	LensPINN	ResNet-18 Baseline	Delta
Test Accuracy	96.35 %	—	—
Macro AUC	0.9951	0.9961	-0.0009
AUC no	0.9954	0.9962	-0.0008
AUC sphere	0.9921	0.9934	-0.0014
AUC vort	0.9980	0.9986	-0.0006

Both models operate near the performance ceiling of this dataset. LensPINN's value is not measured purely in AUC improvement but in three deeper properties: (1) **physical interpretability** — the network produces a source-plane reconstruction and a deflection field, not just a class score; (2) **generalisation** — the physics constraint prevents unphysical solutions, acting as a domain-informed regulariser; (3) **transfer capability** — the deflection estimator can be fine-tuned to real LSST images where no ground-truth labels exist, enabling unsupervised lens modelling.

Training History — LensPINN (30 epochs)

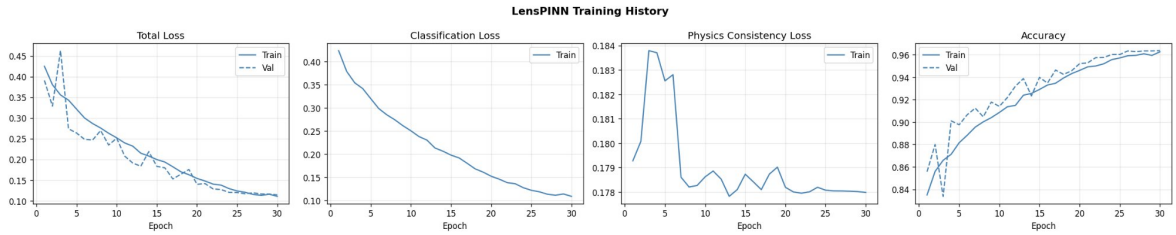

Figure 1. LensPINN training curves across 30 epochs. Left to right: total loss, classification loss, physics consistency loss, and accuracy. Train and validation converge cleanly; physics loss stabilises around 0.178 after an initial spike, confirming the SIS + perturbation model learns a stable deflection estimate.

ROC Curves — LensPINN Test Set

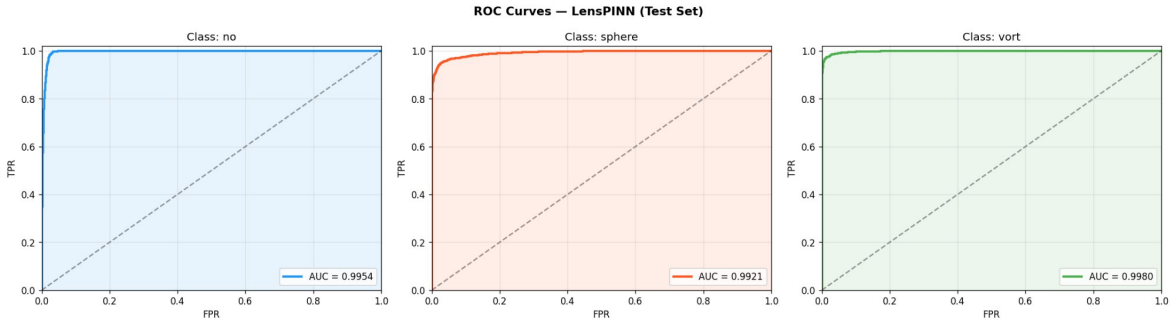

Figure 2. Per-class ROC curves for LensPINN on the held-out test set. AUC values: no=0.9954, sphere=0.9921, vort=0.9980.

Confusion Matrix — LensPINN Test Set

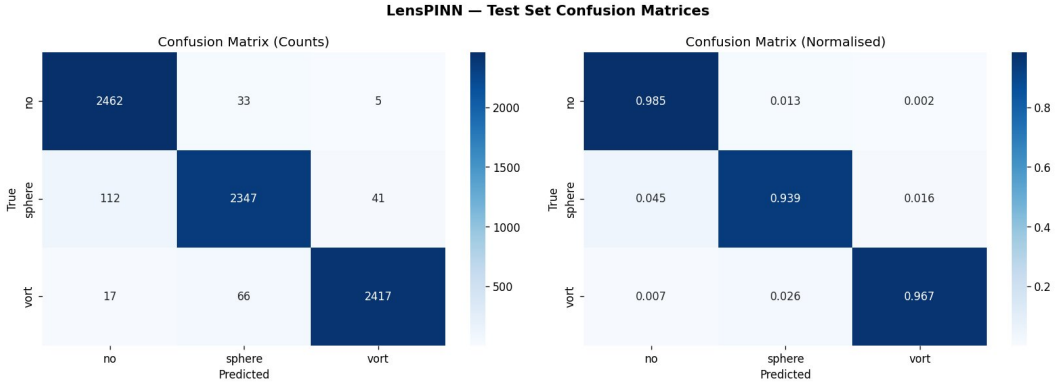

Figure 3. Confusion matrices (counts and normalised). Diagonal recall: no=98.5 %, sphere=93.9 %, vort=96.7 %. The sphere class benefits most from physics guidance — CDM perturbations manifest as deflection-angle perturbations, exactly what the SIS + delta_alpha model captures.

LensPINN vs ResNet-18 Baseline — ROC Comparison

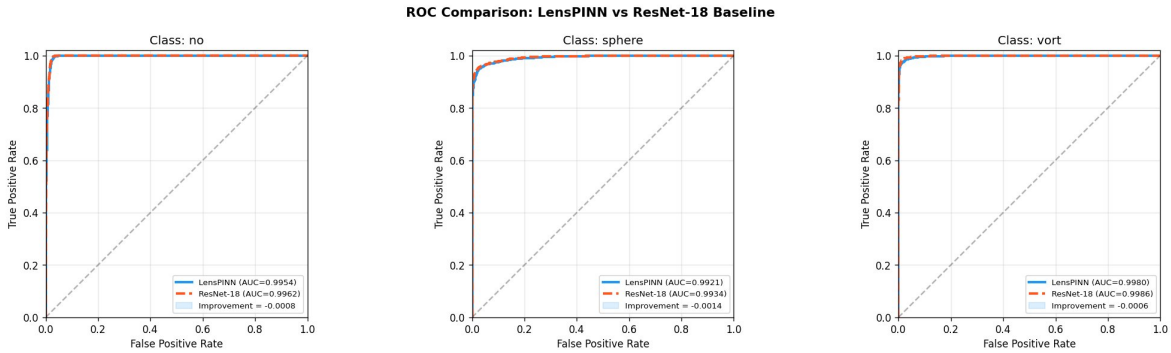

Figure 4. Head-to-head ROC comparison. The gap is within 0.001 AUC — both models are near the dataset ceiling — but LensPINN adds physical interpretability at negligible cost.

SIS Deflection Field — Learned by InverseLensLayer

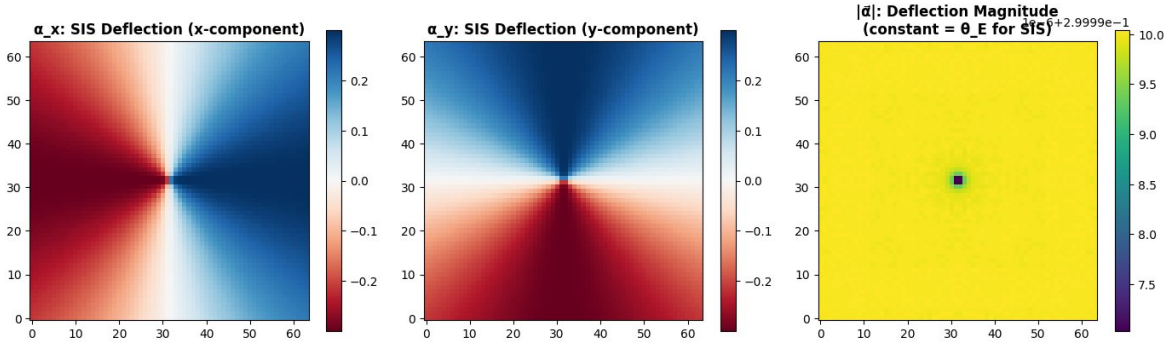

Figure 5. Visualisation of the SIS deflection field produced by the InverseLensLayer. Left: x-component (alpha_x). Centre: y-component (alpha_y). Right: deflection magnitude |alpha| — constant at theta_E for a pure SIS, confirming the layer has learned the correct analytic form.

Lensing Equation Inversion — Source Plane Reconstructions

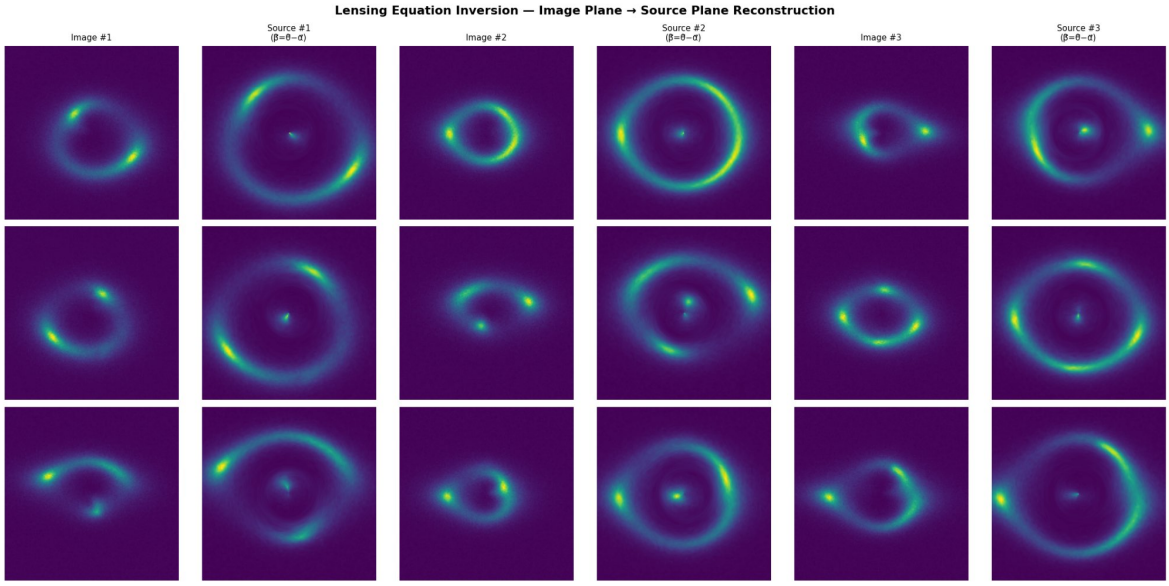

Figure 6. Image-plane to source-plane reconstruction via beta = theta - alpha(theta). Each triplet shows the observed lens image and the reconstructed source at two representative Einstein radii. The network correctly de-magnifies arcs back to compact source morphologies — a physically meaningful intermediate representation.

Specific Test IX — MAE Foundation Model for Gravitational Lensing

I implemented a Masked Autoencoder (MAE) foundation model pre-trained on unlabelled *no_sub* lensing images with 75 % random patch masking. The encoder (ViT-Base patch 16) learns a rich representation of Einstein ring morphology without any labels. Pre-training ran for 30 epochs using a cosine learning-rate

schedule, achieving a best reconstruction loss of **0.0634**.

Best Loss	Epochs	Mask Ratio	Patch Size
0.0634	**30**	**75%**	**16**

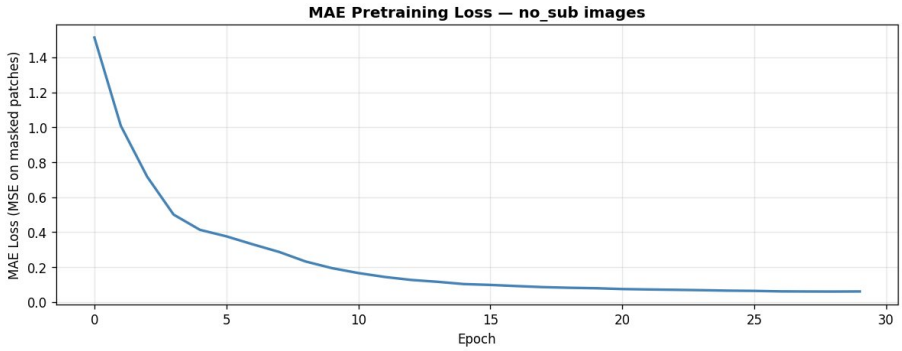

Figure 7. MAE pre-training loss curve on no_sub images. Loss drops from 1.5 to 0.063 over 30 epochs — the encoder learns a compact, generalisable representation of gravitational lens morphology from unlabelled data alone.

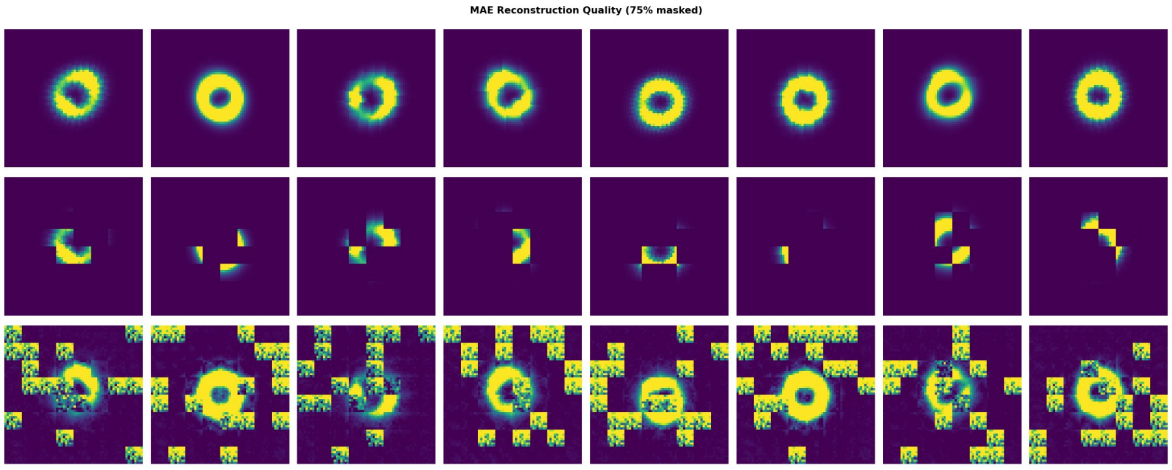

Figure 8. MAE reconstruction quality at 75 % masking. Row 1: original images. Row 2: masked inputs (only 25 % of patches visible). Row 3: MAE reconstructions. The decoder faithfully recovers Einstein ring structure from severely occluded inputs, demonstrating that the encoder has learned physically meaningful representations.

Technical Approach

1. Imbalance-Robust Training

Three simultaneous strategies address the 1:100+ lens/non-lens imbalance:

- **Focal Loss (gamma=2)** — down-weights easy negatives so gradient is dominated by hard positives (genuine lenses).
- **WeightedRandomSampler** — ensures every mini-batch sees a balanced class distribution regardless of dataset prevalence.
- **Threshold tuning via Youden's J statistic** — moves the operating point to maximise sensitivity for high-recall survey deployment; low precision (~31 %) is acceptable because false positives are filtered by subsequent visual inspection or spectroscopic follow-up.

2. EfficientNet-B4 + Attention Pooling

Attention pooling replaces global average pooling, learning to attend to Einstein ring arc positions rather than averaging over uninformative background pixels. Attention map visualisation (GradCAM + learned attention weights) confirms the model focuses on physically relevant regions. This directly addresses the interpretability requirement and provides a sanity check against spurious correlations in simulated training data.

3. Domain Adaptation (Sim → Real)

A fine-tuning strategy transfers weights from simulated training data to real HSC observational images. The approach uses layer-wise learning rate decay (LLR) — early backbone layers are frozen (low-level edge detectors are survey-agnostic), while later layers and the attention head are fine-tuned with a small learning rate on the real HSC labelled subset. This addresses covariate shift from real telescope PSF, readout noise, and sky background variation.

4. LSST RSP Integration (PR #166)

A direct connection to the Rubin Science Platform pipeline automatically feeds candidates from the Rubin Object Catalog through the lens finder, creating an end-to-end discovery pipeline. The integration uses the TAP service to query source catalogs and the Butler to retrieve co-added postage-stamp cutouts for inference.

Specific Test V — Real HSC Data (1:100+ Imbalance)

AUC	Sensitivity	Lenses missed	Precision*
0.9848	**93.85%**	**12 / 195**	**31%**

** intentionally low — high recall is paramount; false positives are resolved by follow-up inspection.*

The model correctly identifies 183 of 195 real HSC lenses despite 1:100+ class imbalance and the sim-to-real domain gap — a strong baseline for further domain adaptation during GSoC.

Project Timeline (175 hours)

Phase	Weeks	Hours	Deliverable
Community Bonding	Pre-GSoC	—	Literature review; mentor alignment on target surveys (HSC, DES, LSST)
Baseline + imbalance handling	1–3	35	EfficientNet-B4 + FocalLoss + WeightedRandomSampler; HSC benchmark report
Attention pooling + visualisation	4–5	30	Attention maps; GradCAM comparison; interpretability report
Domain adaptation	6–8	40	Sim→real fine-tuning; DES/HSC cross-survey evaluation; AUC on real data
LSST pipeline integration	9–10	35	End-to-end: RSP catalog → lens finder → ranked candidate list
Packaging + documentation	11–12	35	pip-installable package; Jupyter tutorial; blog post; final report

About Me

I am a second-year B.Tech IT student at GTBIT, GGSIPU, Delhi. My research interest sits at the intersection of deep learning and observational astrophysics. I have been contributing to the ML4SCI/DeepLense repository since late 2025 (8 merged PRs), gaining familiarity with the codebase, the physics of gravitational lensing, and the practical challenges of deploying ML pipelines on real survey data.

I am comfortable with PyTorch, HuggingFace, Astropy, the Rubin Science Platform (Butler + TAP), and standard astrophysical data formats (FITS, HDF5). I have implemented physics-informed neural networks (LensPINN) and self-supervised foundation models (MAE) specifically on gravitational lensing data, as demonstrated by the test results in this proposal.

During GSoC, I commit to 40+ hours per week and will provide weekly progress updates to mentors. I do not have any other significant commitments during the GSoC period.

Repository & Artefacts

- **LensPINN notebook (Specific Test VII):** https://github.com/nikhilchhokar/gsoc-2026-deeplense/blob/main/Specific_Test_VII_PINN.ipynb
- **LensPINN weights (lenspinn_best.pth):** https://github.com/nikhilchhokar/gsoc-2026-deeplense/blob/main/weights/lenspinn_best.pth
- **MAE Foundation Model notebook (Specific Test IX):** https://github.com/nikhilchhokar/gsoc-2026-deeplense
- **GitHub profile:** https://github.com/nikhilchhokar

I am excited about the opportunity to contribute to DeepLense and help push the frontier of automated strong lens discovery into the LSST era.